

Find the Errors

Each of the following programs or program segments has errors. Find as many as you can.

- ```
43. char str[] = "Stop";
 if (isupper(str) == "STOP")
 exit(0);

44. char numeric[5];
 int x = 123;
 numeric = atoi(x);

45. char string1[] = "Billy";
 char string2[] = " Bob Jones";
 strcat(string1, string2);

46. char x = 'a', y = 'a';
 if (strcmp(x, y) == 0)
 exit(0);
```

## Programming Challenges

### 1. String Length

Write a function that returns an integer and accepts a pointer to a C-string as an argument. The function should count the number of characters in the string and return that number. Demonstrate the function in a simple program that asks the user to input a string, passes it to the function, and then displays the function's return value.

### 2. Backward String

Write a function that accepts a pointer to a C-string as an argument and displays its contents backward. For instance, if the string argument is "Gravity" the function should display "ytivarg". Demonstrate the function in a program that asks the user to input a string and then passes it to the function.

### 3. Word Counter

Write a function that accepts a pointer to a C-string as an argument and returns the number of words contained in the string. For instance, if the string argument is "Four score and seven years ago" the function should return the number 6. Demonstrate the function in a program that asks the user to input a string and then passes it to the function. The number of words in the string should be displayed on the screen. *Optional Exercise:* Write an overloaded version of this function that accepts a string class object as its argument.

### 4. Average Number of Letters

Modify the program you wrote for Problem 3 (Word Counter), so it also displays the average number of letters in each word.

### 5. Sentence Capitalizer

Write a function that accepts a pointer to a C-string as an argument and capitalizes the first character of each sentence in the string. For instance, if the string argument is "hello. my name is Joe. what is your name?" the function should manipulate the string so it contains "Hello. My name is Joe. What is your name?"



VideoNote  
Solving the  
Backward  
String  
Problem

Demonstrate the function in a program that asks the user to input a string and then passes it to the function. The modified string should be displayed on the screen. *Optional Exercise:* Write an overloaded version of this function that accepts a string class object as its argument.

## 6. Vowels and Consonants

Write a function that accepts a pointer to a C-string as its argument. The function should count the number of vowels appearing in the string and return that number.

Write another function that accepts a pointer to a C-string as its argument. This function should count the number of consonants appearing in the string and return that number.

Demonstrate these two functions in a program that performs the following steps:

1. The user is asked to enter a string.
2. The program displays the following menu:
  - A) Count the number of vowels in the string
  - B) Count the number of consonants in the string
  - C) Count both the vowels and consonants in the string
  - D) Enter another string
  - E) Exit the program
3. The program performs the operation selected by the user and repeats until the user selects E to exit the program.

## 7. Name Arranger

Write a program that asks for the user's first, middle, and last names. The names should be stored in three different character arrays. The program should then store, in a fourth array, the name arranged in the following manner: the last name followed by a comma and a space, followed by the first name and a space, followed by the middle name. For example, if the user entered "Carol Lynn Smith", it should store "Smith, Carol Lynn" in the fourth array. Display the contents of the fourth array on the screen.

## 8. Sum of Digits in a String

Write a program that asks the user to enter a series of single digit numbers with nothing separating them. Read the input as a C-string or a `string` object. The program should display the sum of all the single-digit numbers in the string. For example, if the user enters 2514, the program should display 12, which is the sum of 2, 5, 1, and 4. The program should also display the highest and lowest digits in the string.

## 9. Most Frequent Character

Write a function that accepts either a pointer to a C-string, or a `string` object, as its argument. The function should return the character that appears most frequently in the string. Demonstrate the function in a complete program.

## 10. replaceSubstring Function

Write a function named `replaceSubstring`. The function should accept three C-string or `string` object arguments. Let's call them *string1*, *string2*, and *string3*. It should search *string1* for all occurrences of *string2*. When it finds an occurrence of

`string2`, it should replace it with `string3`. For example, suppose the three arguments have the following values:

```
string1: "the dog jumped over the fence"
string2: "the"
string3: "that"
```

With these three arguments, the function would return a `string` object with the value "that dog jumped over that fence." Demonstrate the function in a complete program.

### 11. Case Manipulator

Write a program with three functions: `upper`, `lower`, and `reverse`. The `upper` function should accept a pointer to a C-string as an argument. It should step through each character in the string, converting it to uppercase. The `lower` function, too, should accept a pointer to a C-string as an argument. It should step through each character in the string, converting it to lowercase. Like `upper` and `lower`, `reverse` should also accept a pointer to a string. As it steps through the string, it should test each character to determine whether it is upper- or lowercase. If a character is uppercase, it should be converted to lowercase. Likewise, if a character is lowercase, it should be converted to uppercase.

Test the functions by asking for a string in function `main`, then passing it to them in the following order: `reverse`, `lower`, and `upper`.

### 12. Password Verifier

Imagine you are developing a software package that requires users to enter their own passwords. Your software requires that users' passwords meet the following criteria:

- The password should be at least six characters long.
- The password should contain at least one uppercase and at least one lowercase letter.
- The password should have at least one digit.

Write a program that asks for a password and then verifies that it meets the stated criteria. If it doesn't, the program should display a message telling the user why.

### 13. Date Printer

Write a program that reads a string from the user containing a date in the form `mm/dd/yyyy`. It should print the date in the form `March 12, 2014`.

### 14. Word Separator

Write a program that accepts as input a sentence in which all of the words are run together, but the first character of each word is uppercase. Convert the sentence to a string in which the words are separated by spaces and only the first word starts with an uppercase letter. For example the string `"StopAndSmellTheRoses."` would be converted to `"Stop and smell the roses."`

### 15. Character Analysis

If you have downloaded this book's source code from the companion Web site, you will find a file named `text.txt` in the Chapter 10 folder. (The companion Web site is at [www.pearsonhighered.com/gaddis](http://www.pearsonhighered.com/gaddis).) Write a program that reads the file's contents and determines the following:

- The number of uppercase letters in the file
- The number of lowercase letters in the file
- The number of digits in the file

### 16. Pig Latin

Write a program that reads a sentence as input and converts each word to “Pig Latin.” In one version, to convert a word to Pig Latin you remove the first letter and place that letter at the end of the word. Then you append the string “ay” to the word. Here is an example:

English:        I SLEPT MOST OF THE NIGHT  
Pig Latin:     IAY LEPTSAY OSTMAY FOAY HETAY IGHYNAY

### 17. Morse Code Converter

Morse code is a code where each letter of the English alphabet, each digit, and various punctuation characters are represented by a series of dots and dashes. Table 10-9 shows part of the code.

Write a program that asks the user to enter a string, and then converts that string to Morse code.

**Table 10-9** Morse Code

| Character     | Code         | Character | Code   | Character | Code  | Character | Code  |
|---------------|--------------|-----------|--------|-----------|-------|-----------|-------|
| space         | <i>space</i> | 6         | -....  | G         | ---.  | Q         | ---.- |
| comma         | ---..--      | 7         | ---... | H         | ....  | R         | .-.   |
| period        | ..-.-        | 8         | -----  | I         | ..    | S         | ...   |
| question mark | ..-.-.-      | 9         | -----. | J         | .---- | T         | -     |
| 0             | -----        | A         | .-     | K         | -.-   | U         | ...-  |
| 1             | .----        | B         | -...   | L         | .-... | V         | ...-  |
| 2             | ..----       | C         | -.-.   | M         | --    | W         | .---  |
| 3             | ...----      | D         | -..    | N         | -.    | X         | -...- |
| 4             | ....-        | E         | .      | O         | ---   | Y         | -.--- |
| 5             | .....        | F         | ..-.   | P         | .-.-. | Z         | --..  |

### 18. Phone Number List

Write a program that has an array of at least 10 `string` objects that hold people’s names and phone numbers. You may make up your own strings, or use the following:

"Alejandra Cruz, 555-1223"  
"Joe Looney, 555-0097"  
"Geri Palmer, 555-8787"  
"Li Chen, 555-1212"  
"Holly Gaddis, 555-8878"  
"Sam Wiggins, 555-0998"  
"Bob Kain, 555-8712"  
"Tim Haynes, 555-7676"  
"Warren Gaddis, 555-9037"  
"Jean James, 555-4939"  
"Ron Palmer, 555-2783"

The program should ask the user to enter a name or partial name to search for in the array. Any entries in the array that match the string entered should be displayed. For example, if the user enters “Palmer” the program should display the following names from the list:

```
Geri Palmer, 555-8787
Ron Palmer, 555-2783
```

### 19. Check Writer

Write a program that displays a simulated paycheck. The program should ask the user to enter the date, the payee’s name, and the amount of the check (up to \$10,000). It should then display a simulated check with the dollar amount spelled out, as shown here:

Date: 11/24/2014

Pay to the Order of: John Phillips \$1920.85

One thousand nine hundred twenty and 85 cents

Be sure to format the numeric value of the check in fixed-point notation with two decimal places of precision. Be sure the decimal place always displays, even when the number is zero or has no fractional part. Use either C-strings or `string` class objects in this program.

*Input Validation: Do not accept negative dollar amounts, or amounts over \$10,000.*